LAB WORKSHEET

CIDR Subnetting Lab

Work out any subnet from an address in CIDR notation, one step at a time.

The method (example: 192.168.10.75/27)

Step	How	Example
1. Subnet mask	/n = n ones from the left	/27 → 255.255.255.224
2. Host bits	32 − prefix	32 − 27 = 5
3. Block size	256 − mask value in the interesting octet	256 − 224 = 32 (4th octet)
4. Network address	the multiple of the block size at or below the address	64, 96 … 75 sits in 64 → 192.168.10.64
5. Broadcast	next network − 1	96 − 1 → 192.168.10.95
6. Usable range	network + 1 to broadcast − 1	.65 – .94
7. Usable hosts	$2^{\text{host bits}} - 2$	$2^5 - 2 = 30$

Mask values

Bits on in octet	1	2	3	4	5	6	7	8
Mask value	128	192	224	240	248	252	254	255
Block size	128	64	32	16	8	4	2	1

The interesting octet is where the network bits stop. For /17–/23 it's the 3rd octet; for /9–/15 the 2nd. Octets to its right are 0 in the network address and 255 in the broadcast.

Practice — fill in

Address	Mask	Network	Broadcast	Usable range	Hosts
10.4.7.200/26					
172.16.45.9/20					
192.168.1.130/29					
10.77.3.1/12					

Answers: 255.255.255.192 · .192 · .255 · .193–.254 · 62 | 255.255.240.0 · 172.16.32.0 · 172.16.47.255 · .32.1–.47.254 · 4094 | 255.255.255.248 · .128 · .135 · .129–.134 · 6 | 255.240.0.0 · 10.64.0.0 · 10.79.255.255 · 10.64.0.1–10.79.255.254 · 1,048,574